

CODETANTRA

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4.1.1. Pandas - series creation and manipulation03:14

Write a Python program that takes a list of numbers from the user, creates a Pandas series from it, and then calculates the mean of even and odd numbers separately using the `groupby` and `mean()` operations.

Hint: You can group the Series based on whether each number is even or odd.

Input Format:

- The user should enter a list of numbers separated by space when prompted.

Output Format:

- The program should display the mean of even and odd numbers separately.
- Each mean value should be displayed with a label indicating whether it corresponds to even or odd numbers.

Refer to the sample test cases for better understanding regarding input and output format.

Sample Test Cases

seriesMa...

```
1 import pandas as pd
2
3 # Take inputs from the user to create a list of numbers
4 numbers = list(map(int, input().split()))
5
6 # Create a Pandas series from the list of numbers
7 s = pd.Series(numbers)
8 # Grouping by even and odd numbers and calculating the mean
9 grouped = s.groupby(s % 2 == 0).mean()
10
11 # Display the mean of even and odd numbers with labels
12 grouped.index = ['Even' if is_even else 'Odd' for is_even in
13                 grouped.index]
14 print("Mean of even and odd numbers:")
15 print(grouped)
```

TerminalTest cases

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4.1.2. Dictionary to dataframe04:59

A dictionary of lists has been provided to you in the editor. Create a DataFrame from the dictionary of lists and perform the listed operations, then display the DataFrame before and after each manipulation.

Create the DataFrame:

Convert the dictionary to a Pandas DataFrame.

Add a new row:

Take inputs from the user for the new row data (name, age).
Add the new row to the DataFrame.
Display the DataFrame after adding the new row.

Modify a row:

Modify a specific row by changing the age. Take the row index and new age value from the user.
Display the DataFrame after modifying the row.

Delete a row:

Take the row index to be deleted from the user.
Remove the specified row.
Display the DataFrame after deleting the row.

Add a new column:

Add a column Gender with values taken from the user.
Display the DataFrame after adding the new column.

Modify a column:

Convert names to uppercase.
Display the DataFrame after modifying the column.

Delete a column:

Remove the Age column.

datafram...

```
1  '''import pandas as pd
2
3  # Provided dictionary of lists
4  data = {
5      'Name': ['Alice', 'Bob', 'Charlie'],
6      'Age': [25, 30, 35],
7  }
8
9  # Convert the dictionary to a DataFrame
10 df = pd.DataFrame(data)
11
12 # Display the original DataFrame
13 print("Original DataFrame:")
14 print(df)
15
16 # Adding a new row
17
18
19 # Display the DataFrame after adding a new row
20 print("After adding a row:\n",df)
21
22 # Modifying a row
23
24
25 # Display the DataFrame after modifying a row
26 print("After modifying a row:")
27 print(df)
28
29 # Deleting a row
--
```

TerminalTest cases

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4.1.3. Student Information

01:04

Write a program to read a text file containing student information (name, age, and grade) using Pandas. Perform the following tasks:

- Display the first five rows of the data frame.
- Calculate the average age of the students (limit the average age up to 2 decimal places).
- Filter out the students who have a grade above a certain threshold (consider the threshold grade is 'B').

Note:

Refer to the displayed test cases for better understanding.

Sample Test Cases

+

studentin...

SUBMIT

```
1 import pandas as pd
2
3 # Read the text file into a DataFrame
4 file = input()
5 data = pd.read_csv(file, sep="\s+", header=None, names=["Name", "Age",
6 "Grade"])
7
8 # write your code here..
9 print("First five rows:")
10 print(data.head())
11
12 avg_age = round(data["Age"].mean(), 2)
13 print(f"Average age: {avg_age}")
14
15 print("Students with a grade up to B")
16
17 filtered_student = data[data["Grade"] <= "B"]
18 print(filtered_student)
19
20
21
22
```

Terminal

Test cases

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NEXT >

4.2.1. Month with the Highest Total Sales

02:01

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the columns: Date, Product, Quantity, Price, and City.
- Group the data by Month and calculate the total sales for each month.
- Find the month with the highest total sales and display it.
- Also, display the total sales for the best month.

Sample Data:

```
Date,Product,Quantity,Price,City
2025-01-01,Product A,5,20,New York
2025-01-01,Product B,3,15,Los Angeles
2025-01-02,Product A,7,20,New York
2025-01-02,Product C,4,30,Chicago
2025-01-03,Product B,2,15,Chicago
2025-01-03,Product A,8,20,Los Angeles
2025-01-04,Product C,6,30,New York
2025-01-04,Product B,5,15,Los Angeles
2025-01-05,Product A,3,20,Chicago
2025-01-05,Product C,10,30,Los Angeles
```

Note:

The data cannot be displayed in the file. You can refer to the sample data provided for insights.

Sample Test Cases

+

Explorer

monthFor...

SUBMIT

```
1 # import pandas as pd
2
3 ## Prompt the user for the file name
4 file_name = input()
5
6 ## Load the data
7 df = pd.read_csv(file_name)
8
9
10 ## Find the month with the highest total sales
11 best_month =
12 highest_sales =
13
14 print(f"Best month: {best_month}")
15 print(f"Total sales: ${highest_sales:.2f}")
16
17 # import pandas as pd
18
19 ## Prompt the user for the file name
20 file_name = input()
21
22 ## Load the data
23 df = pd.read_csv(file_name)
24
25 ## Find the month with the highest total sales
26 best_month =
27 highest_sales =
28
29 print(f"Best month: {best_month}")
```

Terminal

Test cases

< PREV RESET SUBMIT NEXT >

4.2.2. Best Selling Product

01:12

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the columns: Date, Product, Quantity, Price, and City.
- Find the product that sold the most in terms of quantity sold.
- Display the product that sold the most and the total quantity sold for that product.

Sample Data:

```
Date,Product,Quantity,Price,City
2025-01-01,Product A,5,20,New York
2025-01-01,Product B,3,15,Los Angeles
2025-01-02,Product A,7,20,New York
2025-01-02,Product C,4,30,Chicago
2025-01-03,Product B,2,15,Chicago
2025-01-03,Product A,8,20,Los Angeles
2025-01-04,Product C,6,30,New York
2025-01-04,Product B,5,15,Los Angeles
2025-01-05,Product A,3,20,Chicago
2025-01-05,Product C,10,30,Los Angeles
```

Note:

The data cannot be displayed in the file. You can refer to the sample data provided for insights.

Sample Test Cases

+

Explorer

monthFor...

SUBMIT

Debugger

```
1 # import pandas as pd
2
3 ## Prompt the user for the file name
4 file_name = input()
5
6 ## Load the data
7 df = pd.read_csv(file_name)
8
9 product_sales = df.groupby('Product')['Quantity'].sum()
10 ## Find the product with the highest total quantity sold
11 best_product = product_sales.idxmax()
12 highest_quantity = product_sales.max()
13
14
15
16 ## Display the result
17 print(f"Best selling product: {best_product}")
18 print(f"Total quantity sold: {highest_quantity}")
19 import pandas as pd
20
21 ## Prompt the user for the file name
22 file_name = input()
23
24 ## Load the data
25 df = pd.read_csv(file_name)
26
27 product_sales = df.groupby('Product')['Quantity'].sum()
28 ## Find the product with the highest total quantity sold
29 best_product = product_sales.idxmax()
```

Terminal Test cases

< PREV RESET SUBMIT NEXT >

4.2.3. City that Sold the Most Products

00:37

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the columns: Date, Product, Quantity, Price, and City.
- Group the data by City and calculate the total quantity of products sold for each city.
- Find the city that sold the most products (based on the total quantity sold).

Sample Data:

```
Date,Product,Quantity,Price,City
2025-01-01,Product A,5,20,New York
2025-01-01,Product B,3,15,Los Angeles
2025-01-02,Product A,7,20,New York
2025-01-02,Product C,4,30,Chicago
2025-01-03,Product B,2,15,Chicago
2025-01-03,Product A,8,20,Los Angeles
2025-01-04,Product C,6,30,New York
2025-01-04,Product B,5,15,Los Angeles
2025-01-05,Product A,3,20,Chicago
2025-01-05,Product C,10,30,Los Angeles
```

Note:

The data cannot be displayed in the file. You can refer to the sample data provided for insights.

Sample Test Cases

+

Explorer

monthFor...

SUBMIT

```
1 import pandas as pd
2
3 # Prompt the user for the file name
4 file_name = input()
5
6 # Load the data
7 df = pd.read_csv(file_name)
8
9 # write the code..
10 product_sales=df.groupby('City')['Quantity'].sum()
11 best_city = product_sales.idxmax()
12 # Display the result
13 print(f"City sold the most products: {best_city}")
14
```

Terminal

Test cases

< PREV RESET SUBMIT NEXT >

4.2.4. Most Frequently Sold Product Pairs

01:41



Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the following columns: Date, Product, Quantity, Price, and City.
- For each date, find all pairs of products that were sold together (i.e., two products sold on the same date).
- Output the product pair/s that was sold most frequently.

Sample Data:

```
Date,Product,Quantity,Price,City
2025-01-01,Product A,5,20,New York
2025-01-01,Product B,3,15,Los Angeles
2025-01-02,Product A,7,20,New York
2025-01-02,Product C,4,30,Chicago
2025-01-03,Product B,2,15,Chicago
2025-01-03,Product A,8,20,Los Angeles
2025-01-04,Product C,6,30,New York
2025-01-04,Product B,5,15,Los Angeles
2025-01-05,Product A,3,20,Chicago
2025-01-05,Product C,10,30,Los Angeles
```

Explanation:

Transactions:

- 2025-01-01: Product A, Product B
- 2025-01-02: Product A, Product C
- 2025-01-03: Product B, Product A
- 2025-01-04: Product C, Product B
- 2025-01-05: Product A, Product C

Now, let's count how often the pairs of products appear together:

- **Product A and Product B:** Appear in transactions on 2025-01-01 and 2025-01-03.
- **Product A and Product C:** Appear in transactions on 2025-01-02 and 2025-01-05.
- **Product B and Product C:** Appears in transactions on 2025-01-04.

Explorer

frequentl...

SUBMIT

```
1 # import pandas as pd
2 # from itertools import combinations
3 # from collections import Counter
4
5 ## Prompt user to input the file name
6 # file_name = input()
7
8 ## Read data from the specified CSV file
9 # df = pd.read_csv(file_name)
10
11 ## write the code
12
13
14 ## Output the most frequent product pairs
15 # import pandas as pd
16 # from itertools import combinations
17 # from collections import Counter
18
19 ## Prompt user to input the file name
20 # file_name = input()
21
22 ## Read data from the specified CSV file
23 # df = pd.read_csv(file_name)
24
25 ## write the code
26
27
28 ## Output the most frequent product pairs
29 # import pandas as pd
30
```

Terminal Test cases

< PREV RESET SUBMIT NEXT >

4.2.5. Titanic Dataset Analysis and Data Cleaning

06:12



You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset. For each question, perform necessary data cleaning, transformations, and calculations as required.

1. Display the first 5 rows of the dataset.
2. Display the last 5 rows of the dataset.
3. Get the shape of the dataset (number of rows and columns).
4. Get a summary of the dataset (using `.info()`).
5. Get basic statistics (mean, standard deviation, etc.) of the dataset using `.describe()`.
6. Check for missing values and display the count of missing values for each column.
7. Fill missing values in the 'Age' column with the median age.
8. Fill missing values in the 'Embarked' column with the most frequent value (`mode()`).
9. Drop the 'Cabin' column due to many missing values.
10. Create a new column, 'FamilySize' by adding the 'SibSp' and 'Parch' columns.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2, 3101282,7.925,,S
```

titanicDat...

```
1 # import pandas as pd
2 # import numpy as np
3
4 ## Load the Titanic dataset
5 # data = pd.read_csv('Titanic-Dataset.csv')
6
7 ## 1. Display the first 5 rows of the dataset
8
9
10 ## 2. Display the last 5 rows of the dataset
11
12
13 ## 3. Get the shape of the dataset
14
15
16 ## 4. Get a summary of the dataset (info)
17
18
19 ## 5. Get basic statistics of the dataset
20
21
22 ## 6. Check for missing values
23
24
25 ## 7. Fill missing values in the 'Age' column with the median age
26
27
28 ## 8. Fill missing values in the 'Embarked' column with the mode
29
30
```

Terminal Test cases

< PREV RESET SUBMIT NEXT >

4.2.6. Titanic Dataset Analysis and Data Cleaning - 2

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

1. Create a new column 'IsAlone' which is 1 if the passenger is alone (FamilySize = 0), otherwise 0.
2. Convert the 'Sex' column to numeric values (male: 0, female: 1).
3. One-hot encode the 'Embarked' column, dropping the first category.
4. Get the mean age of passengers.
5. Get the median fare of passengers.
6. Get the number of passengers by class.
7. Get the number of passengers by gender.
8. Get the number of passengers by survival status.
9. Calculate the survival rate of passengers.
10. Calculate the survival rate by gender.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
```

```
1 # import pandas as pd
2 # import numpy as np
3
4 ## Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6 data['FamilySize'] = data['SibSp'] + data['Parch']
7
8 ## 1. Create a new column 'IsAlone' (1 if alone, 0 otherwise)
9
10
11 ## 2. Convert 'Sex' to numeric (male: 0, female: 1)
12
13
14 ## 3. One-hot encode the 'Embarked' column
15
16
17 ## 4. Get the mean age of passengers
18
19
20
21 ## 5. Get the median fare of passengers
22
23
24
25 ## 6. Get the number of passengers by class
26
27
28
29 ## 7. Get the number of passengers by gender
30
```

4.2.7. Titanic Dataset Analysis and Data Cleaning - 3

01:07

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

- Calculate the survival rate by class.
- Calculate the survival rate by embarkation location (Embarked_S).
- Calculate the survival rate by family size (FamilySize).
- Calculate the survival rate by being alone (IsAlone).
- Get the average fare by passenger class (Pclass).
- Get the average age by passenger class (Pclass).
- Get the average age by survival status (Survived).
- Get the average fare by survival status (Survived).
- Get the number of survivors by class (Pclass).
- Get the number of non-survivors by class (Pclass).

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
```

titanicDat...

1

import pandas as pd

2

import numpy as np

3

4

Load the Titanic dataset

5

data = pd.read_csv('Titanic-Dataset.csv')

6

data['FamilySize'] = data['SibSp'] + data['Parch']

7

data['IsAlone'] = np.where(data['FamilySize'] > 0, 0, 1)

8

data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

9

10

1. Calculate the survival rate by class

11

12

13

2. Calculate the survival rate by embarked location

14

15

16

3. Calculate the survival rate by family size

17

18

19

4. Calculate the survival rate by being alone

20

21

22

5. Get the average fare by class

23

24

25

6. Get the average age by class

26

27

28

7. Get the average age by survival status

29

--

Terminal

Test cases

4.2.8. Titanic Dataset Analysis and Data Cleaning - 4

01:18

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

1. Get the number of survivors by gender (Sex).
2. Get the number of non-survivors by gender (Sex).
3. Get the number of survivors by embarkation location (Embarked_S).
4. Get the number of non-survivors by embarkation location (Embarked_S).
5. Calculate the percentage of children (Age < 18) who survived.
6. Calculate the percentage of adults (Age >= 18) who survived.
7. Get the median age of survivors.
8. Get the median age of non-survivors.
9. Get the median fare of survivors.
10. Get the median fare of non-survivors.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
```

titanicDat...

1

import pandas as pd

2

import numpy as np

3

4

Load the Titanic dataset

5

data = pd.read_csv('Titanic-Dataset.csv')

6

data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

7

8

9

1. Get the number of survivors by gender

10

11

12

2. Get the number of non-survivors by gender

13

14

15

3. Get the number of survivors by embarked location

16

17

18

4. Get the number of non-survivors by embarked location

19

20

21

5. Calculate the percentage of children (Age < 18) who survived

22

23

24

6. Calculate the percentage of adults (Age >= 18) who survived

25

26

27

7. Get the median age of survivors

28

29

Terminal

Test cases

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